

10.7.90

EE25BTECH11042 - Nipun Dasari

Question:

A circle touches the line $y = x$ at a point P such that $OP = 4\sqrt{2}$, where O is the origin. The circle contains the point $(-10, 2)$ in its interior and the length of its chord on the line $x + y = 0$ is $6\sqrt{2}$. Determine the equation of the circle.

Solution:

The tangent line is $y = x$, or $x - y = 0$. A point $\mathbf{P}$ on this line can be represented parametrically.

$$\mathbf{P} = k \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.1)$$

The distance from the origin is given as $\|\mathbf{P}\| = 4\sqrt{2}$.

$$\begin{aligned} \|\mathbf{P}\| &= |k| \left\| \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\| = |k| \sqrt{2} = 4\sqrt{2} \\ \implies |k| &= 4 \end{aligned} \quad (0.2)$$

This gives two possible points of tangency, $\mathbf{P}_1 = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{P}_2 = -4 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Let the center of the circle be $\mathbf{C} = \begin{pmatrix} h \\ k \end{pmatrix}$. The point of tangency $\mathbf{P}$ is the foot of the perpendicular from the center $\mathbf{C}$ to the line $\mathbf{n}^T \mathbf{x} = c$, where $\mathbf{n} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $c = 0$. The formula is:

$$\mathbf{P} = \mathbf{C} - \frac{\mathbf{n}^T \mathbf{C} - c}{\|\mathbf{n}\|^2} \mathbf{n} \quad (0.3)$$

Substituting the values:

$$\begin{aligned} \mathbf{P} &= \begin{pmatrix} h \\ k \end{pmatrix} - \frac{\begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} - 0}{1^2 + (-1)^2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} h \\ k \end{pmatrix} - \frac{h - k}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} h - \frac{h-k}{2} \\ k + \frac{h-k}{2} \end{pmatrix} = \begin{pmatrix} \frac{h+k}{2} \\ \frac{h+k}{2} \end{pmatrix} = \frac{h+k}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned} \quad (0.4)$$

From the two possibilities for $\mathbf{P}$, we have two cases.

$$\text{Case 1: } \frac{h+k}{2} = 4 \implies h+k = 8 \quad (0.5)$$

$$\text{Case 2: } \frac{h+k}{2} = -4 \implies h+k = -8 \quad (0.6)$$

The circle has a chord of length $6\sqrt{2}$ on the line $x+y=0$. The distance from the center $\mathbf{C}$ to this line is d_{chord} . The half-length of the chord is $L = 3\sqrt{2}$.

$$d_{\text{chord}} = \frac{\|h+k\|}{\sqrt{1^2+1^2}} = \frac{\|h+k\|}{\sqrt{2}} \quad (0.7)$$

The radius r is related by $r^2 = d_{\text{chord}}^2 + L^2$.

$$r^2 = \left(\frac{\|h+k\|}{\sqrt{2}} \right)^2 + (3\sqrt{2})^2 = \frac{(h+k)^2}{2} + 18 \quad (0.8)$$

The radius is also the distance from the center to the point of tangency, $r = \|\mathbf{C} - \mathbf{P}\|$.

$$r^2 = \left\| \begin{pmatrix} h \\ k \end{pmatrix} - \frac{h+k}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\|^2 = \left\| \frac{1}{2} \begin{pmatrix} h-k \\ k-h \end{pmatrix} \right\|^2 = \frac{(h-k)^2}{2} \quad (0.9)$$

Equating the two expressions for r^2 :

$$\begin{aligned} \frac{(h-k)^2}{2} &= \frac{(h+k)^2}{2} + 18 \\ (h-k)^2 &= (h+k)^2 + 36 \\ h^2 - 2hk + k^2 &= h^2 + 2hk + k^2 + 36 \\ -4hk &= 36 \implies hk = -9 \end{aligned} \quad (0.10)$$

We now solve for the center in each case.

For Case 1, $h+k=8$ and $hk=-9$.

$$h(8-h) = -9 \implies h^2 - 8h - 9 = 0 \implies (h-9)(h+1) = 0 \quad (0.11)$$

The centers are $\mathbf{C}_A = \begin{pmatrix} 9 \\ -1 \end{pmatrix}$ and $\mathbf{C}_B = \begin{pmatrix} -1 \\ 9 \end{pmatrix}$.

For Case 2, $h+k=-8$ and $hk=-9$.

$$h(-8-h) = -9 \implies h^2 + 8h - 9 = 0 \implies (h+9)(h-1) = 0 \quad (0.12)$$

The centers are $\mathbf{C}_C = \begin{pmatrix} 1 \\ -9 \end{pmatrix}$ and $\mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

For any of these centers, the radius squared is $r^2 = \frac{(h-k)^2}{2} = \frac{(\pm 10)^2}{2} = 50$.

The point $\mathbf{Q} = \begin{pmatrix} -10 \\ 2 \end{pmatrix}$ is in the circle's interior, so $\|\mathbf{Q} - \mathbf{C}\|^2 < r^2 = 50$.

$$\text{For } \mathbf{C}_A = \begin{pmatrix} 9 \\ -1 \end{pmatrix} : \left\| \begin{pmatrix} -19 \\ 3 \end{pmatrix} \right\|^2 = 370 > 50 \quad (0.13)$$

$$\text{For } \mathbf{C}_B = \begin{pmatrix} -1 \\ 9 \end{pmatrix} : \left\| \begin{pmatrix} -9 \\ -7 \end{pmatrix} \right\|^2 = 130 > 50 \quad (0.14)$$

$$\text{For } \mathbf{C}_C = \begin{pmatrix} 1 \\ -9 \end{pmatrix} : \left\| \begin{pmatrix} -11 \\ 11 \end{pmatrix} \right\|^2 = 242 > 50 \quad (0.15)$$

$$\text{For } \mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix} : \left\| \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\|^2 = 2 < 50 \quad (0.16)$$

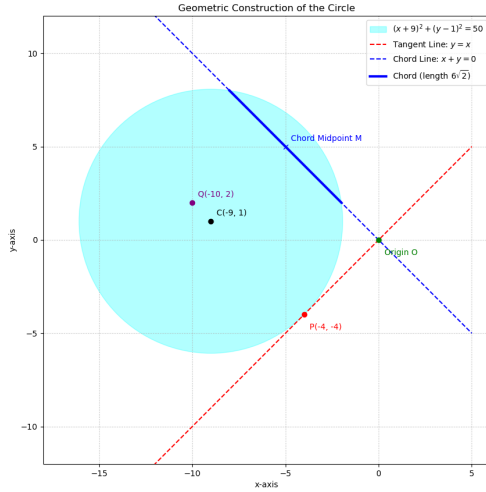
The correct center is $\mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

$$(x + 9)^2 + (y - 1)^2 = 50 \quad (0.17)$$

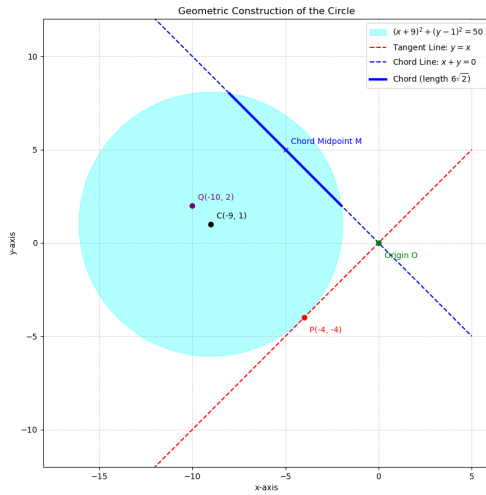
$$x^2 + y^2 + 18x - 2y + 32 = 0 \quad (0.18)$$

In the matrix form $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} 9 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 32 = 0 \quad (0.19)$$



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